Database ER Diagram

Document: E-Numerak — Entity Relationship Diagram (core data model) **Service Provider:** AL MERAK TAX CONSULTANT L.L.C **Date:** 2026-06-20

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1. In simple words

This diagram shows **what information the system stores and how it is linked**.

- A **Company** (the seller/business) has **users**, **customers**, **products**, and the **invoices** it sends.
- Each **Invoice** belongs to one company and one customer, and has many **line items**, an **audit trail**, and an optional **fraud check**.
- For invoices **received** from other businesses, a **Supplier** sends an **Inbound Invoice** to a receiving company.

Only the **core** tables are shown for clarity (the platform has more supporting tables for auth tokens, reporting, etc.).

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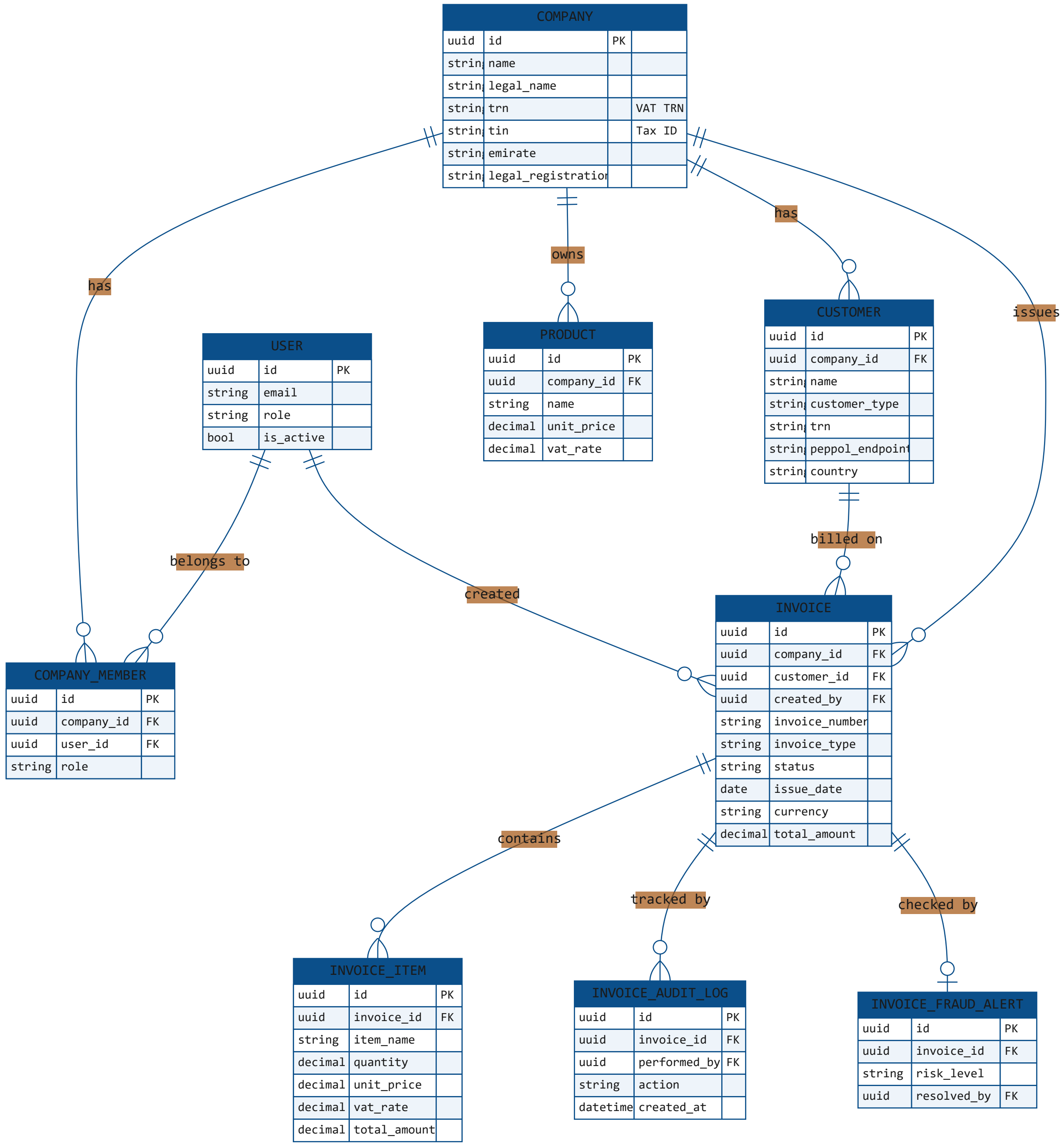
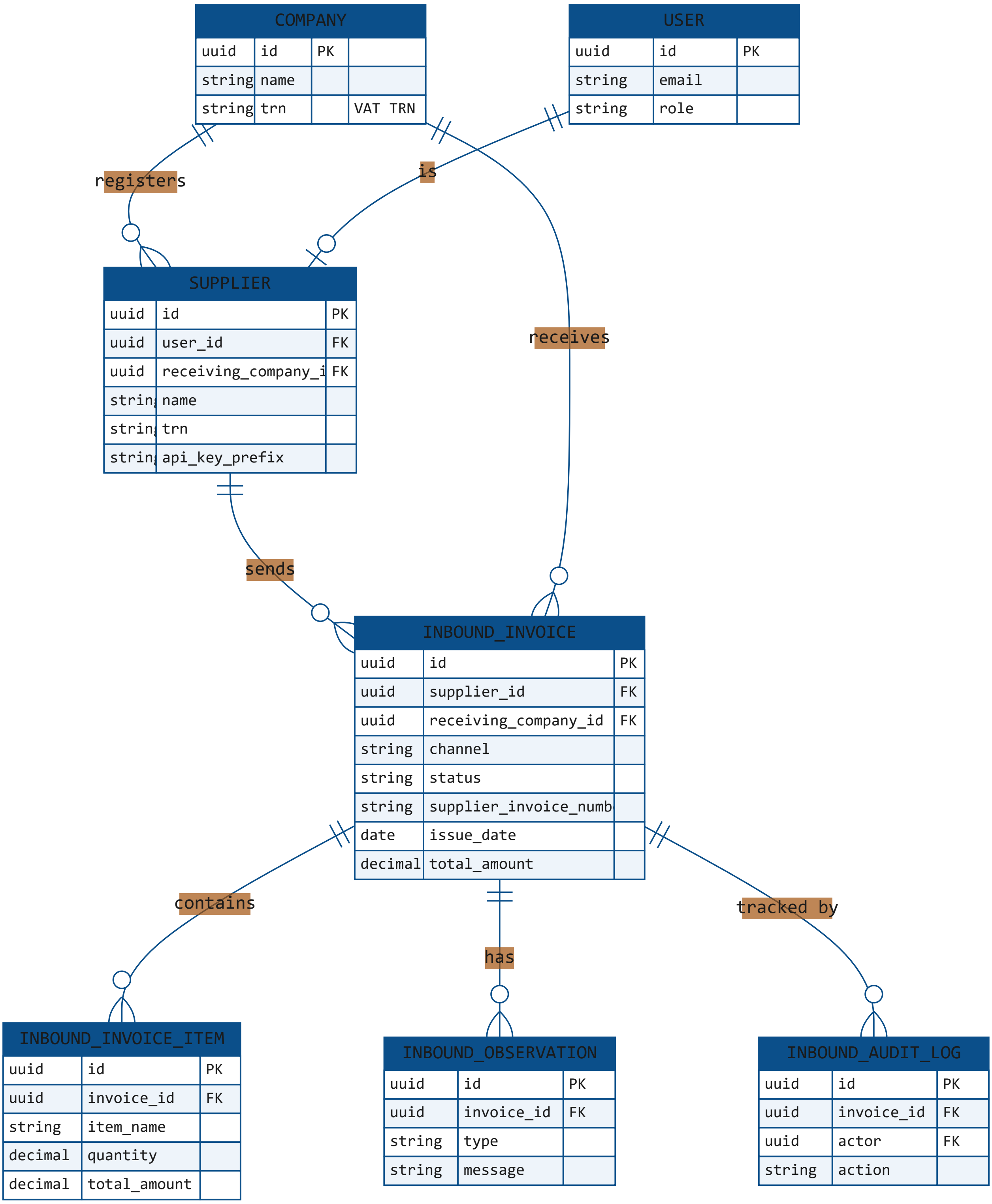
2. Identity & Outbound (Sales) data model

How users, companies, customers, products and **outgoing invoices** relate.



3. Inbound (Purchase) data model

How **suppliers** and **received invoices** relate to the receiving company.


COMPANY
uuid id PK
string name
string trn VAT TRN
USER
uuid id PK
string email
string role
registers
is
receives
SUPPLIER
uuid id PK
uuid user_id FK
uuid receiving_company_i FK
strin name
strin trn
strin api_key_prefix
sends
INBOUND_INVOICE
uuid id PK
uuid supplier_id FK
uuid receiving_company_id FK
string channel
string status
string supplier_invoice_numb
date issue_date
decimal total_amount
contains
has
tracked by
INBOUND_INVOICE_ITEM
uuid id PK
uuid invoice_id FK
string item_name
decimal quantity
decimal total_amount
INBOUND_OBSERVATION
uuid id PK
uuid invoice_id FK
string type
string message
INBOUND_AUDIT_LOG
uuid id PK
uuid invoice_id FK
uuid actor FK
string action

4. Relationship legend

Symbol	Meaning
`||--o{`	one-to-many (e.g. one Company → many Invoices)
`||--o|`	one-to-one / optional (e.g. one Invoice → at most one Fraud Alert)
`PK`	primary key (UUID)
`FK`	foreign key (link to another table)

5. Key relationships in plain terms

- **User** ↔ **Company** is many-to-many through **Company Member** (a user can belong to several companies; a company can have several users, each with a role).
- **Outbound side:** Company → Customer → **Invoice** → Invoice Items (+ audit log + optional fraud alert).
- **Inbound side:** a **Supplier** (linked to a login user) sends an **Inbound Invoice** to a receiving Company, with its own items, observations and audit log.
- All primary keys are **UUIDs**; money fields use exact **decimal** values (no floating-point rounding).